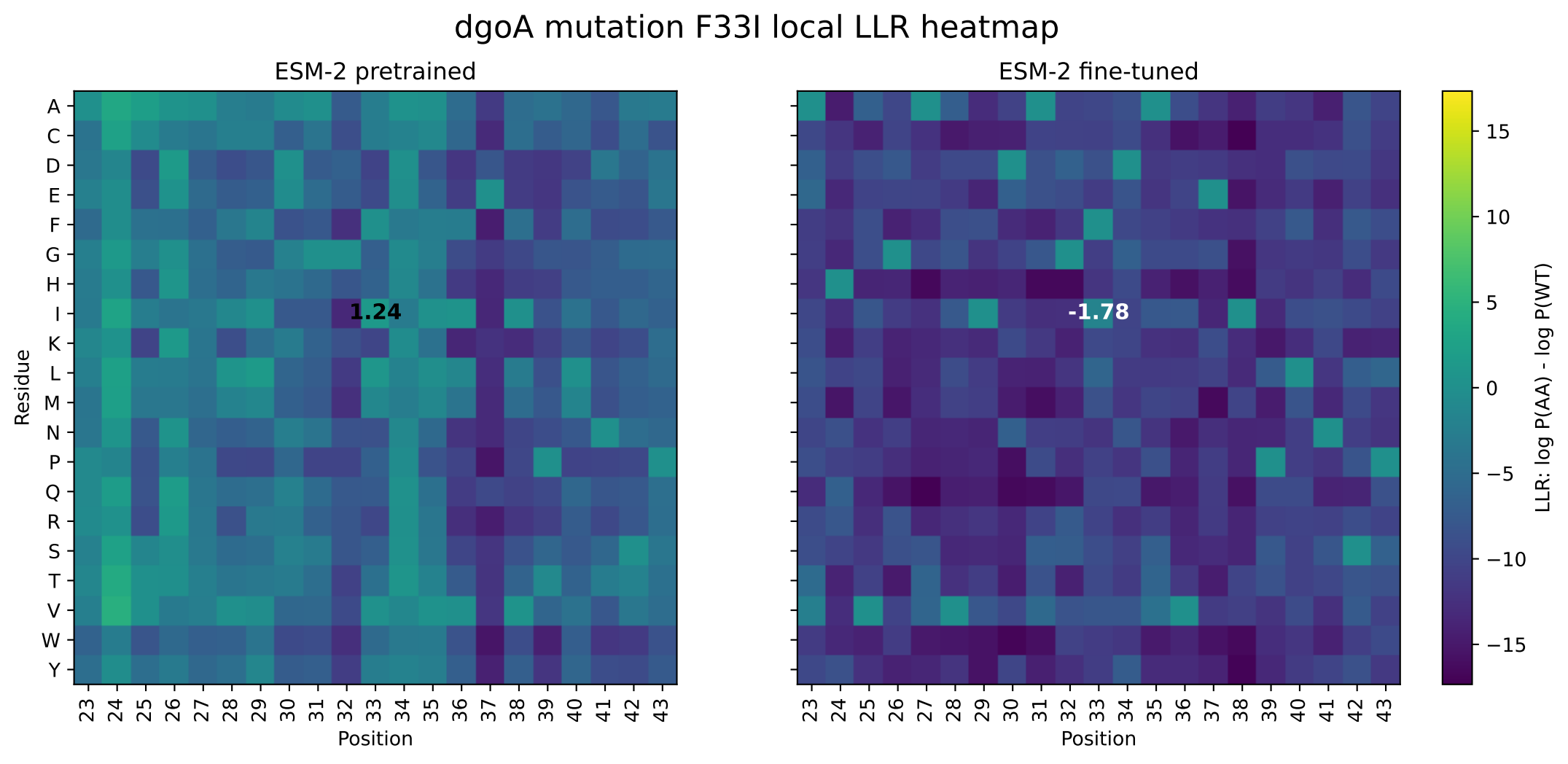
dgoA mutation F33I local LLR heatmap
ESM-2 pretrained
ESM-2 fine-tuned
1.24
-1.78
Residue
A
C
D
E
F
G
H
I
K
L
M
N
P
Q
R
S
T
V
W
Y
23
24
25
26
27
28
29
30
31
32
33
34
35
36
37
38
39
40
41
42
43
Position
15
10
5
0
−5
−10
−15
LLR: log P(AA) - log P(WT)